

Critical Design Review: Advanced Vehicle Control Unit Architecture for Commercial Delta-Footprint 3-Wheelers

1. Executive Summary

The electrification of the commercial auto-rickshaw sector through the implementation of independent rear permanent magnet synchronous hub motors (PMSMs) introduces a profound paradigm shift in vehicle dynamics and control theory. The removal of the traditional mechanical differential eliminates the inherent, passive torque equalization that governs longitudinal traction and lateral stability. Consequently, the entire burden of vehicle stabilization, yaw moment generation, torque vectoring, and traction control is transferred directly to the software and hardware of the Vehicle Control Unit (VCU).¹

The proposed highly advanced VCU architecture represents a sophisticated, physics-based approach designed to manage the unique dynamic properties of a delta-footprint (one front wheel, two rear wheels: 1F2R) vehicle. This architecture is tasked with operating under exceptionally severe constraints: radical gross vehicle weight (GVW) variations ranging from an unloaded 400 kg to a fully loaded 900 kg, operation in harsh commercial environments with

rough roads and steep gradients, and frequent encounters with μ -split surfaces where traction varies drastically between the left and right rear wheels.¹ Furthermore, the system must achieve these objectives relying solely on commercial-grade sensors—specifically a steering angle sensor, a 6-axis Micro-Electro-Mechanical Systems (MEMS) Inertial Measurement Unit (IMU), and wheel speed encoders—while managing the high unsprung mass inherent to direct-drive rear hub motors.¹

The architecture proposes the utilization of a Kinematic-Dynamic Hybrid Observer, a Dual Extended Kalman Filter (DEKF) topology for mass and center of gravity (CG) estimation, a Boundary Layer Saturation Sliding Mode Controller (SMC) for yaw management, and a physics-based torque allocation strategy acting as a virtual differential.⁴ It also incorporates an Active Short Circuit (ASC) hardware failsafe mechanism to comply with the rigorous ASIL D standards defined by ISO 26262.⁷

However, a rigorous and uncompromising Critical Design Review (CDR) reveals fundamental architectural vulnerabilities that compromise the safety, feasibility, and dynamic stability of the vehicle. The mathematical assumptions embedded within the estimation logic and torque allocation strategies exhibit catastrophic interaction failures under limit-handling conditions. The reliance on ASC as a primary failsafe during steep descents violates fundamental thermodynamic limits, and the control-theoretic choices regarding sliding mode control and parameter estimation are severely misaligned with the computational constraints of commercial microcontrollers and the rapid timelines of commercial vehicle development.

Moreover, the entire dynamic framework exhibits a critical oversight regarding the non-minimum phase dynamics inherent to delta-footprint vehicles during reverse driving maneuvers.

This comprehensive report deconstructs the proposed VCU architecture subsystem by subsystem, providing a highly technical analysis of its theoretical underpinnings, exposing the latent hazards within its logic, and delivering strategic directives for necessary architectural rectification.

2. System-Level Context and the Delta-Footprint Dynamic Challenge

Before analyzing the specific algorithms of the VCU, it is imperative to establish the physical realities of the platform. The delta-footprint (1F2R) three-wheeler is fundamentally distinct from standard four-wheeled passenger vehicles and tadpole (2F1R) three-wheelers.

2.1 The Stability Triangle and Load Transfer

In a delta configuration, the vehicle's roll axis is not parallel to the longitudinal centerline. Instead, the stability boundary forms a triangle connecting the single front tire contact patch to the two rear tire contact patches. As the vehicle enters a corner, the lateral acceleration generates a centrifugal force acting through the center of gravity (CG). The resulting load transfer does not merely shift weight from the inner rear wheel to the outer rear wheel; it projects diagonally across the stability triangle.⁹

Because the front of the triangle tapers to a single point, the delta trike is notoriously susceptible to rollover instability, particularly when the CG is located high and biased toward the rear.¹¹ The radical variation in GVW (400 kg to 900 kg) exacerbates this. Commercial auto-rickshaws frequently carry dense cargo or multiple passengers in the rear compartment,

drastically altering the longitudinal CG location (l_f) and the vertical CG height (h). When fully loaded, the static weight distribution may shift from a relatively balanced 40/60 (front/rear) to a heavily biased 20/80 configuration. This extreme rearward bias significantly diminishes the

normal force (F_z) on the single front steering wheel, thereby severely limiting its capacity to generate the lateral tire forces (F_{yf}) required to turn the vehicle and resist yaw moments induced by the rear hub motors.¹³

2.2 The Hub Motor Paradigm

The integration of direct-drive PMSM hub motors introduces massive unsprung weight at the rear axle. While mechanically simplifying the drivetrain by eliminating the transmission, driveshafts, and differential, this high unsprung mass degrades suspension tracking over rough terrain.¹ Consequently, the rear tires will experience high-frequency fluctuations in vertical load

(F_z), leading to transient losses in longitudinal and lateral grip. The VCU's torque allocator must operate with cycle times fast enough to manage these micro-slip events without inducing

longitudinal oscillations or torsional vibration through the chassis.²

3. Kinematic-Dynamic Hybrid Observer and the Dead Reckoning Vulnerability

Accurate vehicle state estimation is the bedrock of advanced vehicle dynamics control. The proposed architecture utilizes a Kinematic-Dynamic Hybrid Observer to synthesize the data from the steering angle sensor, the 6-axis IMU, and the wheel speed encoders to estimate the

vehicle's longitudinal velocity (v_x), lateral velocity (v_y), and yaw rate ($\dot{\psi}$).

3.1 The Algorithmic Fallback Mechanism

The architecture features a specific fallback condition: if the un-driven front wheel locks during panic braking (detected via an unnatural deceleration limit where the wheel speed derivative exceeds physical vehicle deceleration capabilities), the Extended Kalman Filter (EKF) drops the corrupted front-wheel data. In this state, the system falls back to IMU and rear-wheel dead reckoning to maintain the velocity estimate.

3.2 The Reality of MEMS IMU Integration Drift

While this fallback strategy appears logically sound on a macro level, it harbors a fatal mathematical flaw at the implementation level. Commercial-grade MEMS IMUs are plagued by significant stochastic errors, including broadband thermo-mechanical noise (random walk) and low-frequency bias instability (flicker noise).³

In a standard EKF, the IMU accelerations act as the control inputs to propagate the state equations forward in time (the prediction step), while the wheel speeds provide the absolute velocity reference to correct the state estimates (the measurement update step).¹⁸ When the front wheel locks and its data is dropped, and assuming the rear wheels are also undergoing heavy braking slip (rendering their speed data highly unreliable for absolute vehicle velocity), the EKF is effectively starved of its measurement update.

The observer is forced into a state of pure integration (dead reckoning). Integrating a noisy

acceleration signal a_x to find velocity v_x results in a linear accumulation of error over time:

$$v_x(t) = \int_0^t (a_x(\tau) + b_x + \eta) d\tau = v_x(0) + \int_0^t a_x(\tau) d\tau + b_x t + \int_0^t \eta d\tau$$

where b_x is the uncompensated accelerometer bias and η is the white noise.

Without an independent speed reference to bound the estimation error, the velocity state will diverge from reality within a fraction of a second.²⁰ The downstream dynamic bicycle model

relies entirely on v_x to calculate the expected reference yaw rate ($\dot{\psi}_{des} = \frac{v_x}{L+K_{us}v_x^2} \delta_f$).

Feeding a rapidly drifting, erroneously high velocity estimate into the yaw controller will cause the VCU to calculate massive, false yaw errors. The torque allocator will respond by commanding aggressive, destabilizing differential torques to the rear motors during a critical panic braking event, likely inducing a loss of control.

Table 2: Error Propagation in MEMS IMU Dead Reckoning (Without External Update)

Time Elapsed (s)	Velocity Error (m/s)	Position Error (m)	Downstream VCU Impact
0.0 s	0.0	0.0	Nominal operation.
0.1 s	~0.1 - 0.5	~0.05	Yaw reference begins to deviate; minor torque vectoring errors.
0.5 s	~1.0 - 2.5	~0.5 - 1.0	Severe yaw reference distortion; VCU perceives false oversteer/understeer.
1.0+ s	> 3.0 (Divergent)	> 2.0 (Divergent)	Control instability; VCU issues maximum differential torque, inducing spin.

Data represents typical bias instability and random walk for low-cost automotive MEMS IMUs during pure integration.²²

4. The Front-Wheel Lockup Paradox: A Fatal Interaction Failure

The most critical architectural vulnerability discovered during this review is a deterministic interaction failure between the Hybrid Observer's lockup fallback strategy and the Torque Allocator's "Yaw-Moment Boundary" calculation.

4.1 The Yaw-Moment Boundary Logic

The proposed architecture correctly identifies that a vehicle lacking a mechanical differential must utilize its rear hub motors to act as a Virtual Differential. During μ -split acceleration (where one rear wheel is on ice and the other on dry tarmac), applying equal torque results in the low-friction wheel spinning up. Applying torque only to the high-friction wheel generates a massive forward force on one side of the vehicle, creating a severe yaw moment (M_z) that acts to spin the vehicle toward the low-friction side.²

To counter this, the architecture implements a highly advanced "Yaw-Moment Boundary." This logic calculates the maximum restoring lateral capacity of the single front tire (F_{yf_max}). It then actively clips the torque commanded to the gripping rear tire to ensure that the resulting

yaw moment never exceeds the front tire's physical ability to resist it, thereby preventing a spin-out.²⁴ This requires a precise real-time estimation of the front tire's grip.

4.2 The Physics of Tire Saturation

The capacity of a tire to generate lateral force is strictly governed by the friction ellipse (or friction circle). The total resultant force (F_{res}) generated by the tire cannot exceed the product of the normal load (F_z) and the road friction coefficient (μ).²⁴

$$F_{res} = \sqrt{F_x^2 + F_y^2} \leq \mu F_z$$

When a wheel locks during braking, its longitudinal slip ratio (λ) reaches 1.0 (100%). At 100% longitudinal slip, all available friction is consumed in the longitudinal direction (F_x).

Consequently, the tire's capacity to generate lateral force (F_y) drops to exactly zero. A locked front wheel is entirely incapable of steering the vehicle or resisting any externally applied yaw moments.²⁷

4.3 The Catastrophic Algorithmic Loophole

The interaction failure manifests precisely when the front wheel locks during a combined braking and transient turning maneuver.

1. The driver applies panic braking. The un-driven front wheel locks ($\lambda = 1, F_y = 0$).
2. The Hybrid Observer detects the unnatural deceleration limit, flags the front wheel data as corrupted, and drops it from the EKF measurement update, falling back to dead reckoning.¹
3. Crucially, by dropping the wheel speed data, the observer *blinds* the VCU to the physical reality that the wheel is locked. The observer estimates a non-zero forward velocity based on the IMU and continues to operate as if the vehicle is in a nominal dynamic state.
4. The Torque Allocator's Yaw-Moment Boundary algorithm queries the state estimator for the front tire's lateral capacity. Because the estimator does not report a $\lambda = 1$ state, the algorithm calculates a falsely high lateral capacity ($F_{yf_max} > 0$).
5. If a yaw disturbance occurs (e.g., hitting a μ -split patch), the yaw controller requests a corrective differential torque from the rear hub motors.
6. The Torque Allocator approves this differential torque, believing the front tire can act as a stabilizing fulcrum to resist the rotation.
7. The rear motors apply the differential torque, generating a yaw moment. However, because the front tire is actually locked and sliding, it provides zero lateral resistance. The vehicle instantly and violently spins out of control around its rear axle.

The degradation logic designed to maintain continuous state estimation inadvertently masks the most critical physical limit of the vehicle. Dropping locked-wheel data without concurrently

broadcasting a boolean "Zero Front Lateral Capacity" flag to the Torque Allocator represents a fatal flaw. In this state, the VCU ceases to be a stability controller and becomes the primary actuator of a catastrophic spin-out.

5. Dual EKF Mass and CG Estimation: Computational Overload and Convergence Perils

The radical GVW variation from 400 kg to 900 kg in commercial auto-rickshaws necessitates real-time parameter estimation. The architecture proposes a Dual Extended Kalman Filter

(DEKF) topology to continuously estimate the mass (m) and the longitudinal (l_f) and vertical (h) center of gravity positions.²⁸

5.1 The Mathematical Burden of the DEKF

A DEKF operates by running two distinct Extended Kalman Filters simultaneously: a state filter and a parameter filter. At each discrete time step k :

1. The state filter predicts the vehicle states (x_k) using the parameter estimates (θ_{k-1}) from the previous step.
2. The parameter filter predicts the parameters (θ_k) using the state estimates (x_{k-1}) from the previous step.
3. Both filters undergo a measurement update step to correct their predictions based on the incoming sensor data.⁵

The critical bottleneck lies in the linearization process. Because the 3-wheeler vehicle dynamics equations are highly non-linear (involving trigonometric transformations, non-linear Pacejka tire models, and complex load transfer equations), the DEKF requires the computation of Jacobian matrices (partial derivatives of the system equations with respect to the states and parameters) at every single time step.

Evaluating and inverting these Jacobian matrices is an computationally heavy operation scaling at $O(n^3)$, relative to the number of states.³⁰ The low-cost microcontrollers (MCUs) typically deployed in the commercial vehicle sector (e.g., lower-tier NXP or Infineon automotive chips) lack the massive parallel processing power or floating-point unit (FPU) bandwidth required to execute these operations within the strict 10-20 millisecond loop times required for real-time torque vectoring.³² Attempting to run a 6-degree-of-freedom state DEKF and a 3-parameter DEKF concurrently will inevitably result in task overruns, violating real-time execution constraints and causing the entire control loop to crash.

5.2 The Persistence of Excitation Requirement

Beyond computational limits, the DEKF parameter filter suffers from a fundamental observability problem in real-world driving. Parameter estimation algorithms require

"persistence of excitation" to converge.²⁹ To accurately estimate the longitudinal CG (l_f), the vehicle must experience significant lateral acceleration to generate measurable yaw responses.

To estimate the vertical CG (h), the vehicle must experience significant longitudinal and lateral accelerations to induce measurable load transfer.³³

Commercial auto-rickshaws operate extensively in congested, low-speed, stop-and-go traffic characterized by linear trajectories. In these scenarios, the system is fundamentally under-excited. The DEKF parameter covariance will shrink (indicating mathematical confidence), but the estimate will converge onto a local minimum or remain stuck at its initial guess.

The proposed architecture states that the parameter filter will be "frozen once converged." If a driver fully loads the vehicle to 900 kg but then drives smoothly in a straight line for 10 minutes, the DEKF may "converge" and freeze at an erroneous 500 kg estimate. When the driver eventually executes a high-speed emergency lane change, the VCU will calculate torque vectoring commands and rollover prevention thresholds based on a 500 kg mass and a low CG. The resulting torque commands will be severely underdamped for a 900 kg vehicle, resulting in immense overshoot, violent sway, and a high probability of rollover.

5.3 Conflation of Grade and Mass

The architecture attempts to estimate true road angle (θ_{road}) entirely within the EKF to filter out suspension pitch. This is a theoretically flawed objective. When an auto-rickshaw accelerates up a hill, the longitudinal accelerometer measures a combination of the vehicle's

forward acceleration and the component of gravity acting along the sensor's axis ($g \sin \theta$). If the EKF is simultaneously trying to estimate the mass of the vehicle and the grade of the road without an independent, low-frequency tilt sensor (like a dedicated dual-axis inclinometer) or a direct measurement of suspension deflection, the system becomes unobservable.³⁵ The filter cannot mathematically distinguish whether a sluggish longitudinal acceleration response is due to an increased vehicle mass (cargo) or an increased road gradient. This cross-coupling will corrupt both the mass and the gradient estimates simultaneously.

6. Control Theoretic Feasibility: Boundary Layer Sliding Mode Control

To track the target yaw rate (ω_{des}) generated by the Dynamic Bicycle Model, the architecture employs a Boundary Layer Saturation Sliding Mode Controller (SMC). The control law is defined as $u(t) = -K \cdot \text{sat}(s/\Phi)$, where s is the sliding surface, K is the switching gain, and Φ is the boundary layer thickness.⁴

6.1 The Motivation for the Boundary Layer

Traditional Sliding Mode Control is a variable structure control method that forces the system

states to reach and subsequently remain on a user-defined sliding manifold ($s = 0$).³⁷ To achieve this, standard SMC uses a discontinuous switching function: $u(t) = -K \cdot \text{sgn}(s)$. This provides absolute robustness against matched uncertainties, external disturbances, and parameter variations (such as the 500 kg payload variation in this vehicle).

However, the infinite-frequency switching of the $\text{sgn}(s)$ function cannot be realized in physical hardware. Actuator inertia and discrete sampling times cause the system to oscillate rapidly around the sliding surface, a phenomenon known as chattering.⁴ In an architecture utilizing rigid, direct-drive PMSM hub motors with high torque density, chattering is unacceptable. It will excite unmodeled high-frequency structural resonances in the chassis, cause intense torsional vibration, degrade ride comfort, and rapidly overheat the motor inverters.⁴

To mitigate this, the architecture replaces the discontinuous signum function with a continuous saturation function ($\text{sat}(s/\Phi)$), creating a boundary layer of thickness Φ around the sliding surface.⁴¹

6.2 The Illusion of Robustness and Tuning Impossibility

While the boundary layer successfully eliminates chattering, it introduces severe theoretical compromises that render it unsuitable for a commercial 3-wheeler facing extreme parameter variations.

1. **Destruction of Invariance:** The primary advantage of SMC is its robust invariance.

However, inside the boundary layer ($|s| < \Phi$), the controller ceases to be a sliding mode controller. The saturation function acts linearly, and the system degenerates into a high-gain proportional feedback controller.⁴ Consequently, all theoretical guarantees of robustness to parameter variations and external disturbances are lost within this region.⁴

2. **Steady-State Yaw Error:** Because pure sliding mode is never achieved (the state

trajectory evolves within the boundary layer rather than strictly on $s = 0$), the system will exhibit a finite steady-state error.³⁹ In the context of vehicle dynamics, this means the actual yaw rate will constantly deviate slightly from the target yaw rate. When driving on a long, straight μ -split road, this steady-state error manifests as a constant drift, forcing the driver to continuously apply micro-corrections to the steering wheel to maintain a straight line. This directly violates the design requirement for a "natural driving feel".¹

3. **The Gain Scheduling Nightmare:** The switching gain K must be chosen to be strictly greater than the upper bound of the system uncertainties to guarantee reachability. In a vehicle where the mass more than doubles (400 kg to 900 kg) and the road friction varies by an order of magnitude (ice $\mu = 0.1$ to dry asphalt $\mu = 1.0$), the uncertainty bounds are massive.⁴³ If K is set high enough to cover the fully loaded, low-friction

worst-case scenario, the required boundary layer Φ must also be incredibly wide to prevent chattering when the vehicle is unloaded. A wide boundary layer results in sluggish, sloppy yaw response that fails to reject sudden crosswinds. Conversely, if Φ is narrow, the unloaded vehicle will experience violent chattering. Properly tuning K and Φ requires exhaustive, multi-dimensional gain-scheduling across speed, estimated mass, and estimated road friction.³⁸ This exponential calibration burden is completely incompatible with the tight development timelines and cost constraints of commercial auto-rickshaws.

Table 3: Comparative Analysis of Yaw Control Strategies

Control Strategy	Robustness to GVW Variation	Susceptibility to Chattering	Steady-State Error	Calibration Complexity
Standard PID	Very Low	None	High (under load)	Low
Standard SMC	Absolute	Severe (Destructive)	Zero	Medium
Boundary Layer SMC (Proposed)	Poor (inside boundary)	Mitigated	Finite (Drift)	Extremely High
Second-Order SMC (STA) (Recommended)	Absolute	None (Continuous)	Zero	Medium

7. Thermal and Hardware Limits of the Active Short Circuit (ASC) Failsafe

To comply with the strict functional safety requirements of ISO 26262, the architecture features an ASIL D hardware watchdog. If this watchdog detects a hard fault or complete freeze of the VCU software, it physically bypasses the controller and commands an Active Short Circuit (ASC) in the PMSM stators. The intent is to provide a physics-based, solid-state deceleration to a safe stop.⁴⁴

7.1 The Physics of the Active Short Circuit

An ASC is executed by simultaneously turning on all three upper (or all three lower) Insulated-Gate Bipolar Transistors (IGBTs) in the motor inverter. This effectively shorts the three phase windings of the stator together.⁴⁵ As the permanent magnets on the rotor spin past the shorted stator coils, they induce a massive back-electromotive force (BEMF), which in turn drives a large short-circuit current (I_{sc}) through the windings.

At high mechanical speeds, the inductive reactance (ωL_d) dominates the phase resistance (

R_s). The steady-state short-circuit current converges asymptotically to the machine's characteristic current: $I_{sc} = -\Psi_m / L_d$, where Ψ_m is the permanent magnet flux linkage and L_d is the d-axis inductance.⁴⁴

The resulting braking torque profile is highly non-linear. Torque peaks sharply at relatively low speeds where the resistive and inductive terms balance. However, as speed increases, the massive short-circuit current acts almost entirely in the d-axis (the demagnetizing direction), pushing the motor deep into flux weakening. Consequently, the braking torque actually *drops off* at higher speeds.⁴⁵

7.2 The 15% Gradient Thermal Catastrophe

The prompt demands an assessment of this ASC strategy for a fully loaded 900 kg vehicle descending a long 15% gradient. Physics dictates that relying on ASC as a primary sustained deceleration failsafe under these conditions will result in catastrophic hardware destruction.

A 15% gradient corresponds to an angle (θ) of approximately 8.53° . The gravitational force pulling the 900 kg vehicle down the slope is:

$$F_{gravity} = m \cdot g \cdot \sin(\theta) \approx 900 \text{ kg} \cdot 9.81 \text{ m/s}^2 \cdot 0.148 \approx 1306 \text{ Newtons}$$

To prevent the vehicle from accelerating uncontrollably down the mountain, the rear hub motors must generate a continuous opposing braking force of 1306 N. Because ASC braking torque diminishes at higher speeds, the vehicle will likely accelerate until aerodynamic drag and mechanical rolling resistance bridge the gap between the gravitational pull and the motor's high-speed ASC torque limit.

During this prolonged descent, the motor operates as a short-circuited generator. The kinetic energy of the 900 kg mass is entirely converted into electrical energy, which, because the

circuit is shorted, is dissipated exclusively as heat via $I^2 R$ copper losses in the stator windings.⁴⁸

Direct-drive in-wheel hub motors are severely disadvantaged regarding thermal management. Unlike chassis-mounted motors that benefit from dedicated liquid cooling jackets, hub motors are completely enclosed within the wheel shell. The thermal resistance path from the internal stator coils, across the internal air gap, to the external rotor shell, and finally to the ambient air is notoriously high.¹⁵ They rely entirely on passive air cooling over the exterior shell, which is frequently shrouded by the vehicle's bodywork.

Subjecting the hub motors to the continuous, massive I_{sc} required to hold a 900 kg vehicle on a 15% grade will cause a rapid, exponential rise in internal temperatures.⁵²

1. **Insulation Failure:** Standard automotive stator winding insulation (Class F or Class H) is rated for peak operating temperatures of 155°C to 180°C. Exceeding this threshold will melt the resin, causing inter-turn short circuits and explosive phase-to-phase faults within the motor.⁵⁴
2. **Irreversible Demagnetization:** More critically, the Neodymium-Iron-Boron (NdFeB)

permanent magnets mounted on the rotor are highly sensitive to temperature. If the internal temperature exceeds 140°C to 160°C, the magnets will undergo irreversible demagnetization.⁴⁵ Furthermore, the demagnetization threshold is significantly lowered when the magnets are simultaneously subjected to the massive opposing magnetic field generated by the ASC d-axis current.⁴⁵

Once the magnets are demagnetized, the motor loses all ability to generate torque—propulsive or braking. The vehicle will enter a catastrophic free-fall down the gradient. Therefore, while ASC is an excellent *transient* safe state to clamp voltages during a split-second fault before mechanical brakes engage, it is fundamentally unsafe to use as a standalone, solid-state deceleration method for heavy commercial vehicles on steep gradients.

8. Remaining Loopholes: Non-Minimum Phase Reverse Dynamics and Rollover Margins

An exhaustive review of the proposed dynamic models reveals critical hidden edge cases that expose the vehicle to severe dynamic instability, particularly at the opposite ends of the operational spectrum.

8.1 The Reverse-Driving Dynamic Instability

The most glaring loophole in the proposed architecture is the implicit assumption that the vehicle dynamics are symmetric in forward and reverse. They are absolutely not.

The delta footprint (1F2R) is dynamically stable when driving forward because the center of gravity (CG) is located behind the single front steering wheel. The steering mechanism leads the mass. However, when the driver selects reverse gear, the kinematic paradigm violently shifts. The vehicle dynamically transforms into a Rear-Wheel-Steer (RWS) vehicle, analogous to a forklift, where the single steered wheel is now trailing the massive rear-biased CG.⁵⁶

In a commercial 3-wheeler, the static weight distribution is heavily biased towards the rear axle (the leading axle in reverse) to accommodate the cargo box and the heavy hub motors. When driving in reverse, if the driver applies a steering input, the trailing single wheel generates a lateral force. This force creates a yaw moment that swings the heavy leading CG outward, away from the center of the turn. This creates a vicious positive feedback loop: the outward swing increases the slip angle on the trailing wheel, which generates more lateral force, which further accelerates the yaw rate, swinging the CG out even faster.⁵⁸

This phenomenon renders the delta-footprint fundamentally, dynamically unstable in reverse; it exhibits non-minimum phase behavior, meaning the initial vehicle response is in the opposite direction of the steady-state target.⁶⁰ The proposed "Dynamic Bicycle Model" for target yaw

generation assumes forward velocity ($v_x > 0$). If the VCU applies this identical logic while

$v_x < 0$, the state matrix poles shift to the right-half plane. The Sliding Mode Controller, attempting to minimize the yaw error using standard forward-logic torque vectoring, will actively and aggressively drive the vehicle into an uncontrollable spin or jack-knife.⁵⁷ The architecture completely lacks a dedicated Reverse Control Law.

8.2 The Predictive Rollover Algorithm Vulnerability

The proposed architecture features a highly advanced rollover prevention algorithm utilizing a delta-specific load transfer ratio:

$$PLTR_{delta} = \frac{h \cdot (a_y \cos \alpha + a_x \sin \alpha)}{g \cdot d_{tip}} + C_p (\dot{\delta}_f \cdot v_x)$$

This is a fundamentally excellent formulation. Unlike four-wheeled vehicles where lateral and longitudinal load transfers are generally decoupled, the delta trike's triangular footprint means

that braking (a_x) and cornering (a_y) forces combine to push the vehicle's mass over a

diagonal tipping axis (d_{tip}).⁹ Projecting these forces onto this specific axis is mathematically

correct. Furthermore, the inclusion of the predictive term $C_p (\dot{\delta}_f \cdot v_x)$ represents state-of-the-art foresight, allowing the VCU to anticipate impending load transfer based on the driver's steering rate ($\dot{\delta}_f$) before the physical acceleration fully manifests in the IMU chassis readings.

However, the efficacy of this entire equation hinges absolutely on the accuracy of the vertical

CG height variable (h). As established in Section 5.2, the DEKF parameter filter will likely

struggle to converge on a true h value due to a lack of dynamic excitation in commercial stop-and-go driving. If a driver loads 500 kg of cargo high in the rear compartment (drastically raising h), but the DEKF is frozen at an unloaded h estimate, the $PLTR_{delta}$ calculation will severely underestimate the physical load transfer. The calculated threshold will not be breached, the predictive torque intervention will not trigger, and the vehicle will roll over before the VCU recognizes the danger.

8.3 Low-Speed Maneuverability and Hub Motor Cogging

Commercial auto-rickshaws spend a vast majority of their operational life executing low-speed maneuvers (0 - 5 km/h), such as parking, U-turns, and navigating dense traffic. Direct-drive PMSM hub motors, by their electromagnetic nature, exhibit significant cogging torque and spatial torque ripple at these very low rotational speeds.⁶²

The VCU's boundary layer SMC is designed to execute high-frequency micro-corrections via torque vectoring to maintain the target yaw rate. At low speeds, these micro-commands will interact destructively with the inherent electromagnetic torque ripple of the motors. The constructive and destructive interference between the commanded torque and the physical cogging will result in a highly unrefined, shuddering, and jerky physical feedback to the driver and chassis. This fundamentally degrades the required "natural driving feel" and induces unnecessary fatigue on the suspension mounting points.¹

9. ISO 26262 Functional Safety and Degradation

Assessment

The architecture incorporates specific degradation strategies designed to comply with ISO 26262 safety goals.⁷

- **Steering Loss Triggering Yaw Damping:** The proposal to trigger "Yaw Damping" upon the loss of the steering angle sensor is an exceptionally robust, fail-operational strategy.⁶⁴ Without steering angle data, the VCU cannot utilize the Dynamic Bicycle

Model to determine the driver's intended yaw rate (ω_{des}). Attempting to guess intent is highly dangerous. By falling back to a damping mode, the VCU uses the IMU to measure uncommanded vehicle rotation and commands the rear hub motors to apply differential counter-torque strictly to resist rotation.⁶⁵ This effectively acts as an electronic stability anchor, ensuring the vehicle maintains a straight, stable trajectory regardless of road disturbances, allowing the driver to bring the vehicle to a controlled halt using the mechanical brakes.

- **CAN Timeouts Triggering Regen Stops:** Initiating autonomous local inverter regenerative braking upon communication loss between the VCU and MCU is appropriate and standard practice. Furthermore, the explicit requirement that the Electronic Parking Brake (EPB) fires *strictly* at 0 km/h is a critical safety feature. Firing a mechanical EPB dynamically at speed on the rear axle would cause instantaneous, uncontrollable rear-wheel lockup, virtually guaranteeing a violent spin-out. Ensuring the vehicle reaches a standstill via inverter regen before engaging the EPB is correct.

10. Strategic Directives and Architectural Rectifications

Based on the uncompromising findings of this Critical Design Review, the VCU architecture must undergo the following strategic rectifications to bridge the gap between theoretical vehicle dynamics and commercial viability:

Table 4: Identified Failures and Mandated Architectural Corrections

Subsystem	Identified Vulnerability	Mandated Rectification
State Estimation	DEKF imposes an impossible computational load ($O(n^3)$, Jacobians) and risks freezing on false parameters due to a lack of dynamic excitation.	Deprecate DEKF. Implement a Recursive Least Squares (RLS) algorithm with a variable forgetting factor for mass estimation based purely on longitudinal dynamics. Pair this with a standard Unscented Kalman Filter (UKF) for state tracking to eliminate Jacobian computation.

Torque Allocation	VCU commands differential yaw moments during front-wheel lockup, acting on phantom grip and causing catastrophic spin-outs.	Implement a Friction Saturation Interlock. If the front wheel slip exceeds a critical threshold ($\lambda > \quad$), the VCU must instantly attenuate the target yaw moment (M_z) to zero, disabling differential torque until front lateral grip is physically restored.
Yaw Control	Boundary Layer SMC causes pseudo-sliding mode steady-state drift errors and requires impossible multi-dimensional gain scheduling for GVW variations.	Upgrade to Second-Order Sliding Mode (SOSM). Specifically, utilize the Super-Twisting Algorithm (STA). This provides mathematically continuous control (eliminating mechanical chattering) while maintaining absolute robust invariance to the 400–900kg mass variations without requiring a compromised boundary layer.
Hardware Failsafe	ASC will cause thermal destruction of hub motor stators and irreversible magnet demagnetization on long 15% gradient descents.	Reclassify ASC as a Transient Safe State. The system must only use ASC to clamp inverter voltages for < 2 seconds. The architecture must integrate communication with the mechanical friction brake module (via a redundant ABS/ESC hydraulic circuit) to handle sustained kinetic energy dissipation on gradients.
Reverse Dynamics	Forward-driving dynamic model causes violent non-minimum phase instability (jack-knifing) when applied to reverse driving.	Implement a Dedicated Reverse Control Law. The VCU must recognize reverse gear, invert its yaw control logic, heavily dampen high-frequency steering inputs via differential braking, and strictly limit reverse velocity to < 10 km/h to prevent RWS

		instability.
Estimation Fallback	Dead reckoning using pure IMU integration after front wheel data loss diverges instantly, corrupting downstream controllers.	Synthesize a Pseudo-Velocity Reference. Instead of pure open-loop integration, blend un-driven rear-wheel speeds (compensated for estimated slip) or heavily filter the IMU acceleration through a kinematic slip-angle observer to bound the velocity integration drift.

11. Conclusion

The proposed VCU architecture demonstrates a profound and sophisticated understanding of three-wheeled vehicle dynamics. The formulation of the delta-specific predictive rollover equations and the integration of ISO 26262 compliant yaw-damping mechanisms represent state-of-the-art engineering for the commercial auto-rickshaw sector. The theoretical intent to completely replace a mechanical differential with a software-defined torque allocation strategy is highly ambitious and mechanically sound.

However, the architecture requires critical, fundamental modifications before it can be deployed on physical hardware. The mathematical assumption that dropped sensor data can be safely replaced by IMU dead reckoning ignores the physical reality of tire friction limits, creating a fatal algorithmic loop where the VCU initiates the very spin-out it was designed to prevent. Furthermore, the reliance on theoretical continuous control limits—such as infinite thermal capacity for ASC braking on steep gradients, and perfect, rapid parameter convergence for Dual EKF—directly conflicts with the harsh, noisy, and computationally constrained realities of the commercial automotive environment.

Implementing the mandated transitions to Super-Twisting Sliding Mode Control, Recursive Least Squares parameter estimation, and establishing strict operational boundaries for front-tire saturation and reverse RWS dynamics will transform this theoretical architecture into a robust, fail-safe, and production-ready system capable of handling the extreme environments of commercial three-wheeler operation.

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